Title Slide

Topic: Solar Power Forecasting Using Machine Learning

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Course: Computer Engineering Presentation

Introduction & Motivation

- The Shift to Renewable Energy:
- Solar energy is growing rapidly worldwide.
- It is clean, sustainable, and reduces carbon footprint.
- The Core Challenge (Intermittency):
- Solar generation is highly dependent on weather conditions.
- Clouds, rain, and seasonal changes cause sudden power drops.

The Problem for Power Grids

- Grid Stability Requirements:
- Electricity supply and demand must always balance in real time.
- Overproduction or underproduction causes grid instability or blackouts.
- Why Accurate Forecasting Matters:
- Power plants need day-ahead and hour-ahead predictions to manage reserves.

Data Collection & Features

- Input Data Sources:
- Real-time weather APIs (e.g., OpenWeatherMap).
- Historical meteorological records and solar panel telemetry.
- Key Features Used:
- Ambient temperature and humidity.
- Solar irradiance (sunlight intensity).
- Cloud cover percentage and wind speed.

Machine Learning Methodology

- Model Selection:
- Regression models and Time-Series algorithms (e.g., Random Forest, XGBoost).
- Training Process:
- Feeding historical weather parameters into the model.
- Training the algorithm to recognize patterns between weather and power output.(2024). Machine Learning for Renewable Energy.

Evaluation & Expected Benefits

- Performance Metrics:
- Using Root Mean Square Error (RMSE) to measure prediction accuracy.
- Project Benefits:
- Better energy storage management.
- Lower operational costs for smart grids.

References & Conclusion

- Conclusion: Machine learning provides a reliable way to overcome the unpredictability of solar energy.

- References:

- Antonanzas, J., et al. (2016). Review of photovoltaic power forecasting. Solar Energy, 136, 78-111.

- OpenWeatherMap API Documentation. (2026). Current Weather Data and Forecast APIs.

- Towards Data Science. (2024). Machine Learning for Renewable Energy: Predicting Solar Output.